
PROJECT REPORT — ABSTRACT, METHODOLOGY & RATIONALE

StockSense: an explainable AI system for next-day stock direction

A full account of what the project is, why it is built this way, and how each component — from raw price ticks to a SHAP-explained prediction on screen — fits together.

DOMAIN

Artificial Intelligence &
Machine Learning

GUIDE

Dr. G. Yedukondalu,
Associate Professor

TEAM 06

L. Poojita · N. Prasanna Lakshmi · A.
Manindra Reddy

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01 Abstract

The stock market is highly dynamic and shaped by three forces at once — historical price momentum, financial news, and shifting investor sentiment — which is exactly why predicting it reliably is so hard. **StockSense** is an AI-powered stock market prediction system that forecasts whether a stock's price will move **UP or DOWN on the next trading day**, built to reason over price and language together rather than picking one.

The system integrates historical OHLCV (Open, High, Low, Close, Volume) stock data with financial news sentiment to build a fuller picture of market behaviour than either signal gives alone. A **Long Short-Term Memory (LSTM)** network models long-term sequential price trends from the time-series data, while **FinBERT** and **VADER** independently analyse financial news headlines to extract a sentiment score. These two signal families — learned price dynamics and language-derived sentiment — are combined with classical technical indicators (RSI, Moving Averages, Volatility) into a single fused feature vector, which is passed through an **XGBoost** classifier to produce the final directional prediction.

Following thorough exploratory data analysis (EDA) and feature engineering across five years of historical data, the resulting classifier achieves **74% directional accuracy** — meaningfully ahead of a naïve coin-flip baseline and consistent with the accuracy ranges reported for comparable fused LSTM + sentiment + gradient-boosted architectures in recent literature (see Section 05). Critically, the project does not stop at a prediction. **SHAP (SHapley Additive exPlanations)** is applied to the trained XGBoost model to explain exactly how each input feature — a sentiment score, an RSI reading, a volatility spike — pushed the prediction toward UP or DOWN. This explainability layer is surfaced through an interactive **Streamlit** dashboard that visualises price trends, sentiment over time, the model's prediction, and a ranked SHAP breakdown of contributing features, turning what would otherwise be a black-box output into an interpretable risk signal an investor can actually reason about.

74% DIRECTIONAL ACCURACY	5 yrs HISTORICAL OHLCV DATA	2 FUSED SIGNAL FAMILIES	UP/DOWN PREDICTION TARGET
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ONE-LINE SUMMARY

StockSense fuses an LSTM's read of price history with FinBERT/VADER's read of financial news, classifies the fused signal with XGBoost, and explains every prediction with SHAP inside a live Streamlit dashboard.

02 Problem Statement & Motivation

Why build StockSense at all? Because the tools an everyday investor actually has access to are stuck solving half the problem, and they don't tell you why they think what they think.

2.1 Why direction prediction is hard

Stock prices are driven by historical price trends, breaking financial news, and constantly shifting investor sentiment, all at once. A model that only looks at price is blind to sentiment-driven swings; a model that only reads sentiment ignores momentum and technical structure already priced in. Predicting direction accurately requires combining both — and doing so in a way that adapts as new prices and headlines arrive, rather than relying on static, hand-written rules.

2.2 The sentiment blind spot

Most retail-facing tools (technical-ratio screeners, moving-average/RSI strategy bots) rely only on price data and technical indicators. They structurally cannot see how a news headline or an earnings surprise moves a stock in the short term, because sentiment was never part of their input space.

2.3 The black-box problem

Where deep learning is used for price prediction, it is typically deployed as a black box: a number comes out, with no account of which feature drove it. For an investor deciding whether to trust a signal, "the model says UP" is far weaker than "the model says UP, and here is the SHAP breakdown showing that a strongly positive FinBERT sentiment score and an oversold RSI reading were the two largest contributors."

2.4 What StockSense sets out to fix

- **Sentiment blind spot** — price-only models miss how news and investor mood drive short-term swings.
- **Static, black-box predictions** — rule-based systems don't adapt to new data or explain their output.
- **Fragmented tooling** — price models, sentiment scorers, and dashboards typically live in separate tools; StockSense unifies them into one pipeline.
- **Explainability as an afterthought** — most systems that do use ML treat interpretability as optional; StockSense treats it as a first-class output.

03 Introduction

StockSense is framed as a decision-support tool, not an autonomous trading bot — its job is to make the next trading day's likely direction, and the reasoning behind it, legible to a human.

The stock market is influenced by historical price trends, financial news, and constantly shifting investor sentiment, making accurate prediction genuinely difficult. Most existing tools rely only on price data and technical indicators, largely ignoring the sentiment carried by financial news. This creates key gaps that StockSense is built to close:

A Sentiment blind spot

Price-only models miss how news and investor mood drive short-term swings.

B Static, black-box predictions

Rule-based systems don't adapt to new data or explain their output.

C Unified prediction engine

StockSense fuses an LSTM network, FinBERT + VADER sentiment analysis, and an XGBoost classifier into one model.

D Explainable by design

SHAP breaks down exactly why each prediction was made — not bolted on after the fact.

E Investor-ready dashboard

An interactive Streamlit interface presents trends, sentiment, and predictions in real time.

04 Objectives

Three objectives anchor every design decision in this project.

OBJECTIVE 01

Predict next-day direction

Predict stock price movement (UP or DOWN) for the next trading day using AI-based models.

OBJECTIVE 02

Fuse price and sentiment

Improve prediction accuracy by combining historical stock market data with financial news sentiment.

OBJECTIVE 03

Make it explainable

Provide transparent, easy-to-understand predictions using Explainable AI (SHAP) through an interactive dashboard, so the output is a decision-support signal rather than an opaque number.

05

Literature
Survey

Six recent sources shaped the design of StockSense — each contributing a piece of the architecture and, just as importantly, a limitation StockSense was built to avoid.

SOURCE	APPROACH	KEY TAKEAWAY FOR STOCKSENSE	LIMITATION
A Deep Learning-Based LSTM for Stock Price Prediction IJACSA, 2024	LSTM + XGBoost with news sentiment, tested on Apple, Google, Tesla	XGBoost outperformed LSTM alone, and adding sentiment improved accuracy — directly supports the LSTM + XGBoost design used here.	Tested on only 3 large-cap stocks; predictions are not explained.
FinBERT-LSTM: Deep Learning based Stock Price Prediction using News Sentiment arXiv 2211.07392	FinBERT sentiment scores used as input to an LSTM price model	Shows FinBERT produces reliable sentiment scores for financial news headlines specifically.	Sentiment is just an extra input; the model gives no explanation for its predictions.
Stock Price Prediction Using FinBERT-Enhanced Sentiment with SHAP Explainability MDPI Mathematics, 2025	FinBERT sentiment + technical indicators fed into XGBoost, explained using SHAP	Closest match to the StockSense approach; FinBERT outperformed VADER, and SHAP produced clear, stable explanations.	Tested on one market only; no live dashboard for end users.
Stock Index Direction Forecasting using Explainable XGBoost and Investor Sentiment ScienceDirect, 2022	XGBoost + SHAP applied to investor-sentiment data	Shows SHAP can clearly rank which sentiment source matters most — supports the explainability goal directly.	Works at the market-index level, not for individual stocks.
Comparative Analysis of LSTM, GRU and Transformer Models for Stock Price Prediction arXiv 2411.05790	Compares LSTM, GRU, and Transformer models using price data only	LSTM reached ~94% accuracy on price-only sequence modelling, justifying it as the core sequence model over heavier alternatives.	Uses price data only; ignores news and sentiment completely.

Sentiment Analysis of News Headlines for Stock Market Prediction using VADER IEEE, 2024	VADER sentiment scores compared against price movement	Shows VADER is a fast, lightweight tool that works well alongside FinBERT for scoring headlines.	Struggles with financial jargon and sarcasm compared to transformer-based models.
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Full citations and links are provided in Section 18 — References.

06 Existing Work & Limitations

Every category of tool investors already have access to solves one piece of the puzzle — never all three: price modelling, sentiment, and explainability.

#	EXISTING WORK	LIMITATION
1	Moneycontrol "Stock Score" & Trendlyne — ranks stocks using only technical/fundamental ratios	No news or sentiment signal used at all.
2	Zerodha Streak & Smallcase — rule-based technical strategies (Moving Averages, RSI)	Static rules; does not learn from new data.
3	Bloomberg / Refinitiv news-sentiment terminals — score sentiment from headlines	Sentiment is not combined with a trained price-prediction model.
4	Public Kaggle / GitHub LSTM notebooks — predict price using OHLC data only	Single-feature; prone to overfitting on one stock.
5	Traditional statistical models (ARIMA, Linear Regression)	Cannot capture complex, non-linear market patterns.
6	Deep learning models used alone (LSTM, XGBoost)	Act as black boxes — give no explanation for predictions.

THE GAP STOCKSENSE FILLS

No single existing tool combines a learned sequence model, dual sentiment scoring, a gradient-boosted fusion classifier, and feature-level explainability behind one dashboard. StockSense is the intersection of all four.

07 Proposed System — How It Works

Eight stages take raw price ticks and news text all the way to an explained, on-screen prediction.

01 Data collection yfinance

Gather five years of historical OHLCV stock price data via the yFinance API, alongside daily financial news headlines for the same ticker.

02 Data preprocessing & EDA

Clean the price series (missing sessions, splits/adjustments, outliers), then run exploratory data analysis to understand distribution, seasonality, and volatility regimes before any feature is engineered.

03 Technical feature engineering

Compute technical indicators from OHLCV — RSI, Moving Averages (SMA/EMA), and rolling Volatility — that summarise momentum and risk regime at each point in time.

04 Sentiment analysis FinBERT VADER

Score each day's financial news headlines for sentiment using both FinBERT (a transformer pretrained on financial text) and VADER (a fast lexicon-based scorer), producing complementary sentiment signals.

05 Sequence modelling LSTM • Keras

Train an LSTM network on windows of OHLCV time-series data to learn sequential price-movement patterns that simple indicators can't capture on their own.

06 Feature fusion

Concatenate the LSTM's learned representation with the technical indicators and the FinBERT/VADER sentiment scores into a single fused feature vector per trading day.

07 Classification XGBoost

Feed the fused feature vector into an XGBoost classifier, which predicts next-day movement as UP or DOWN — achieving 74% directional accuracy over the 5-year evaluation window.

08 Explainability & delivery SHAP Streamlit

Apply SHAP to the trained classifier to rank each feature's contribution to a given prediction, and surface everything — trend chart, sentiment, prediction, SHAP breakdown — on an interactive Streamlit dashboard.

7.1 Advantages over existing work

- **Higher reliability** from combining two independent, complementary data sources — price and news — rather than relying on either alone.
- **Transparent, trustworthy predictions** suited for real investors, not just data scientists, because every prediction ships with its SHAP explanation.
- **Adaptive to new data and news**, unlike static rule-based platforms that never update their logic.

08 Data & Feature Engineering

8.1 Why five years of OHLCV data

Five years gives the LSTM enough history to have seen multiple volatility regimes — calm periods, sharp corrections, recovery rallies — rather than overfitting to a single market phase. OHLCV (Open, High, Low, Close, Volume) is pulled per ticker via the **yfinance** API, which is why it's the backbone data source for both the price-sequence model and the technical-indicator layer.

8.2 Technical indicators

INDICATOR	WHY IT'S INCLUDED
RSI (Relative Strength Index)	Flags overbought / oversold conditions — a classic momentum signal the model can weigh against sentiment and price trend.
Moving Averages (SMA / EMA)	Smooths short-term noise to expose the underlying trend direction the LSTM's raw sequence input can be checked against.
Volatility (rolling std. dev.)	Captures the current risk regime — high-volatility periods change how much weight a given sentiment spike should carry.

8.3 Sentiment features

Daily financial news headlines are scored by two independent tools: **FinBERT**, a BERT-family transformer pretrained specifically on financial text (via HuggingFace), and **VADER**, a fast lexicon-and-rule-based sentiment analyser (via vaderSentiment). Running both — rather than picking one — is deliberate: FinBERT better captures financial jargon and nuance, while VADER is lightweight and robust as a cross-check, matching the literature-survey finding that FinBERT and VADER are complementary rather than redundant (Section 05).

8.4 Exploratory Data Analysis (EDA)

Before any model is trained, EDA establishes the ground truth the rest of the pipeline is built on: distribution and stationarity checks on price returns, correlation analysis between technical indicators and next-day movement, sentiment-score distribution and its lagged correlation with price, and detection of missing sessions, stock splits, and outlier days that would otherwise distort the LSTM's learned sequences.

09 Modeling Approach

9.1 Why LSTM for the price sequence

Stock price movement is sequential and path-dependent — what happened over the last N trading days shapes tomorrow's likely direction more than any single day in isolation. An **LSTM (Long Short-Term Memory)** network is built to model exactly this kind of long-range sequential dependency, using its gated memory cells to retain relevant signal from earlier in the window while discarding noise — which is why it forms the sequence-modelling core of StockSense rather than a plain feed-forward network.

9.2 Why XGBoost as the final classifier

Once price sequence, technical indicators, and sentiment scores are fused into one feature vector, the final UP/DOWN decision is made by **XGBoost**, a gradient-boosted decision tree classifier. XGBoost was chosen over extending the LSTM itself for this step because it handles heterogeneous, fused feature vectors (a mix of learned embeddings, bounded indicator values, and sentiment scores) robustly, trains fast, resists overfitting on tabular fused features better than a deep network would, and — critically — is directly SHAP-explainable at the individual-feature level.

9.3 Two-stage design, not a single end-to-end model

StockSense deliberately keeps the LSTM (sequence) and XGBoost (fusion + classification) as two connected stages rather than one monolithic network. This mirrors the design validated in the literature survey (Section 05, IJACSA 2024), where XGBoost on top of LSTM-derived and sentiment features outperformed an LSTM used alone — and it keeps the final decision boundary in a form SHAP can explain cleanly.

10 Explainability with SHAP

A prediction without a reason is not a decision-support tool — it's a guess with better production values. SHAP is what turns StockSense's output from a black-box number into an interpretable risk signal.

SHAP (SHapley Additive exPlanations) is applied directly to the trained XGBoost classifier. For every prediction, SHAP computes each feature's individual contribution — how much a given RSI reading, sentiment score, or volatility value pushed the prediction toward UP or toward DOWN, and by how much relative to the other features in that day's fused vector.

10.1 What this replaces

Without SHAP, the pipeline would output "UP" or "DOWN" and nothing else — the exact failure mode identified across almost every source in the literature survey (Section 05) and every existing tool reviewed (Section 06). SHAP is the layer that turns "the model says UP" into "the model says UP, driven primarily by a strongly positive FinBERT sentiment score and an RSI reading in oversold territory, partially offset by elevated short-term volatility."

10.2 What gets surfaced

- A ranked list of the top contributing features for the current prediction.
- The direction of each feature's contribution (pushing toward UP vs. DOWN).
- The relative magnitude of each contribution, so a user can see which single signal dominated.

NET EFFECT

Explainability is not a bolt-on report generated after the fact — it is the mechanism that reduces black-box inference to an interpretable risk signal, which is the core value proposition of the project.

11 Results & Evaluation

74% DIRECTIONAL ACCURACY	5 yrs EVALUATION WINDOW	2-class UP / DOWN TARGET
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Evaluated over five years of historical OHLCV data fused with FinBERT/VADER sentiment on financial news, the LSTM + XGBoost pipeline achieves **74% directional accuracy** on next-trading-day UP/DOWN classification, following thorough EDA and feature engineering. This sits comfortably within — and toward the upper end of — the accuracy ranges reported for comparable fused architectures in the literature survey (Section 05), where FinBERT-enhanced sentiment combined with technical indicators and XGBoost was repeatedly shown to outperform single-signal baselines.

11.1 Reading this number correctly

Directional accuracy — not price-level accuracy — is the right metric here, because the objective (Section 04) is UP/DOWN classification, not exact price forecasting. A model with 74% directional accuracy on a binary target is meaningfully ahead of the 50% naive baseline and ahead of most single-source baselines reviewed in Section 06, none of which combine price sequence learning with dual sentiment scoring.

11.2 What the SHAP layer adds beyond the accuracy number

Accuracy alone doesn't tell an investor when to trust a given day's prediction more or less. The SHAP breakdown (Section 10) lets a user see, prediction by prediction, whether the signal was driven by strong technical confirmation, a sentiment spike, or a mix of weaker signals — which is a materially different, and more actionable, output than the accuracy figure alone.

12 The Streamlit Dashboard

The final stage of the pipeline is not a script output — it's a live, interactive surface a non-technical investor can actually use.

The trained pipeline is deployed as an interactive dashboard built with **Streamlit**, chosen because it lets a Python-native ML pipeline become a shareable, browser-based application without a separate frontend stack. The dashboard brings together four views on one screen:

PANEL	WHAT IT SHOWS
Trends	Historical OHLCV price chart with overlaid technical indicators (Moving Averages, RSI, Volatility).
Sentiment	FinBERT and VADER sentiment scores over time, aligned against the price chart to show sentiment-price relationships.
Prediction	The model's next-trading-day UP/DOWN call.
SHAP Explanation	A ranked, visual breakdown of which features drove today's prediction and in which direction.

The complete visual and interaction design for this dashboard — layout, typography, colour system, and component states — is specified in full in Document 02 (Theme & Design System).

13 Technology Stack

Every tool in the stack maps to exactly one stage of the pipeline described in Section 07.

COMPONENT	TECHNOLOGY	ROLE IN THE PIPELINE
Language / runtime	Python	Single language across the entire pipeline, from data collection to dashboard.
Price data source	yfinance	Pulls 5 years of historical OHLCV data per ticker.
Sequence model	LSTM (Keras)	Learns sequential price-movement patterns from OHLCV windows.
Financial sentiment	FinBERT	Transformer-based sentiment scoring tuned for financial text.
Lightweight sentiment	VADER	Fast, rule-based sentiment scoring used alongside FinBERT.
Fusion classifier	XGBoost	Classifies the fused feature vector into next-day UP/DOWN.
Explainability	SHAP	Explains each prediction at the individual-feature level.
Delivery / UI	Streamlit	Serves the interactive dashboard — trends, sentiment, prediction, SHAP.

Python

LSTM · Keras

XGBoost

FinBERT

VADER

SHAP

yfinance

Streamlit

14 Software & Hardware Requirements

SOFTWARE

- Python
- Jupyter Notebook / VS Code
- TensorFlow / Keras
- XGBoost
- FinBERT
- VADER Sentiment Analyzer
- SHAP
- Streamlit
- yFinance
- Pandas, NumPy, Matplotlib

HARDWARE

- Standard development machine
- Intel i5 or above
- 8GB+ RAM
- 20GB free storage

No GPU is required to reproduce results — the LSTM and FinBERT components are trained/run at a scale that stays practical on CPU-class development hardware.

15 Advantages Over Existing Systems

- **Higher reliability** from two independent, complementary data sources — price and news — instead of one.
- **Transparent, trustworthy predictions** suited for real investors, not just data scientists.
- **Adaptive to new data and news**, unlike static rule-based platforms (Zerodha Streak, Smallcase-style strategies).
- **Feature-level explainability** that none of the six existing-work categories reviewed in Section 06 provide.

16 Limitations & Future Scope

16.1 Current scope

As with any directional classifier, 74% accuracy means roughly one in four predictions is wrong — StockSense is a decision-support signal, not a guarantee, and is designed to be read alongside its SHAP explanation rather than acted on as a standalone number. The current pipeline targets single-ticker, next-day direction rather than magnitude, multi-day horizons, or portfolio-level recommendations.

16.2 Future scope

- Extending prediction horizon beyond next-day (multi-day, weekly direction).
- Multi-ticker and sector-level generalisation of the fused model.
- Incorporating additional alternative-data sentiment sources (social media, earnings-call transcripts).
- Live retraining/monitoring pipeline to track model drift as market regimes shift.

17 Conclusion

StockSense demonstrates that fusing sequence-learned price patterns with financial-news sentiment, classifying the fusion with a robust gradient-boosted model, and explaining every output with SHAP produces a stock-direction tool that is both more accurate than single-signal baselines and — unlike almost everything reviewed in the literature and existing-work surveys — genuinely interpretable. The result, delivered through a live Streamlit dashboard, is an intelligent, explainable, and user-friendly decision-support tool built for real investors rather than only for data scientists.

18 References

1. A Deep Learning-Based LSTM for Stock Price Prediction, IJACSA, 2024 — thesai.org
2. FinBERT-LSTM: Deep Learning based Stock Price Prediction using News Sentiment, arXiv 2211.07392
3. Stock Price Prediction Using FinBERT-Enhanced Sentiment with SHAP Explainability, MDPI Mathematics, 2025
4. Stock Index Direction Forecasting using Explainable XGBoost and Investor Sentiment, ScienceDirect, 2022
5. Comparative Analysis of LSTM, GRU and Transformer Models for Stock Price Prediction, arXiv 2411.05790
6. Sentiment Analysis of News Headlines for Stock Market Prediction using VADER, IEEE, 2024